

Review of Chap IV univariate polynomials

1. FTOA $\forall f(x) \in \mathbb{C}[x], \deg \geq 0$. Then $\exists c \in \mathbb{C}$ s.t. $f(x) = (x-c)g(x)$ for some $g(x) \in \mathbb{C}[x]$. Furthermore $f(x) = a_{n-1} \prod_{i=1}^n (x-c_i)$.

2. For irreducible polynomial in $\mathbb{R}[x]$ are exactly:

① $ax+b, a \neq 0$ ② $ax^2+bx+c, \Delta < 0$

3. Eisenstein's Criterion, let p be a prime,

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \in \mathbb{Z}[x]$$

s.t. ① $p \nmid a_i, 0 \leq i \leq n-1$ ② $p \mid a_n$ ③ $p^2 \nmid a_0$

对非原多项式的加强.

Then $f(x)$ is irreducible in $\mathbb{Z}[x]$ or $\mathbb{Q}[x]$

§1 operations of vector space

Given $V, W/F$ $+, -, \times, \div$

1. $V+W = \{v+w \mid v \in V, w \in W\}$ $\times \rightarrow v \times w$ is meaningful if $v, w \in V$

2. $V \times W = \{(v, w) \mid v \in V, w \in W\}$

$$(V \times W, +, \cdot) \Rightarrow \text{v.s. over } F$$

$$"+": (v, w) + (v', w') = (v+v', w+w')$$

$$"\cdot": \lambda(v, w) = (\lambda v, \lambda w)$$

Pf. $(0, 0)$ is the prime of $V \times W$

$$-(v, w) = (-v, -w) = (-1) \cdot (v, w)$$

Thm. Let V, W be a basis of V, W respectively. Then

$$= \{(v, 0) \mid v \in V\} \cup \{(0, w) \mid w \in W\}$$

$V \times \{0\} \cup \{0\} \times W$ is a basis of $V \times W$

Pf. for simplicity assume $|V|=m$ $|W|=n$

$$\text{Then } v \in V, w \in W \quad v = \sum x_i v_i, \quad w = \sum y_j w_j$$

$$\text{so } (v, w) = (\sum x_i v_i, \sum y_j w_j)$$

$$= (\sum x_i v_i, 0) + (0, \sum y_j w_j)$$

$$= \sum x_i (v_i, 0) + \sum y_j (0, w_j)$$

$$\text{LI: } \sum x_i (v_i, 0) + \sum y_j (0, w_j) = 0$$

$$\Leftrightarrow \sum x_i v_i = 0, \sum y_j w_j = 0$$

cr. Let $\dim V = m, \dim W = n$ Then $\dim V \times W = m+n$

pk. so product of vector space is indeed sum of V 's

Generalize

Given $v_i \mid 1 \leq i \leq n$ Define

$$V_1 \times V_2 \times \dots \times V_n = (V_1 \times V_2 \times \dots \times V_{n-1}) \times V_n$$

$$(V_1 \times V_2) \times V_3 \supseteq V_1 \times (V_2 \times V_3) \Rightarrow \{(v_1, v_2), v_3\} \neq \{v_1, (v_2, v_3)\}$$

Question 1: Discuss $f(x) \in F[x]$ as a function over F ($x \mapsto f(x)$)

Question 2: Try to define a distance over $F[x]$

$$f(x) = \sum_{i=0}^{\infty} a_i x^i \quad g(x) = \sum_{i=0}^{\infty} b_i x^i$$

Review of CH1-II $F \subseteq \mathbb{C}$

Abstract vector space, $(V, +)$ - Abelian group

Def. Let $(V, +)$ be an abelian group with a scalar multiplication

$$F \times V \rightarrow V \quad (1 \cdot v = v \text{ is not natural})$$

Ex. 1. $F^A \rightarrow A^B$

Ex. 2. $(F, +, \cdot)$ are "+" and "." independent?

Kronecker δ is given, others are artificial

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$$

$\cdot +$ is natural

$$(a + b\sqrt{2})(c + d\sqrt{2}) = ac + (b + ad)\sqrt{2}$$

Def (Dimension) V/F , $B = \{v_1, v_2, \dots, v_n\} \subseteq V$, satisfies:

(i) LI (ii) maximal $\rightarrow$ basis

Lemma. Let $B = \{v_1, v_2, \dots, v_n\}$ be basis of V

$$C = \{u_1, u_2, \dots, u_m\} \rightarrow \text{Then } n = m$$

If B be an infinite basis of V , a contradiction is easily found

so Dimension is well-defined

Classification $\begin{cases} \text{finite dimension} \\ \text{infinite dimension: } F[x], F[[x]], \mathbb{R}/\mathbb{Q} \end{cases}$

What should we do next $V \rightarrow v_1, v_2, \dots, v_n, v_{n+1} \in V$
 $V = v_1 \oplus v_2 = v_1/v_2$ decomposition

§ 2. Linear Maps (Linear Transformations)

Given $V, W / \mathbb{F}$, define a map $\sigma: V \rightarrow W$ satisfying:

$$1) \sigma(v + v') = \sigma(v) + \sigma(v') \quad \} \quad \sigma(xv + x'v') = x\sigma(v) + x'\sigma(v')$$

$$2) \sigma(\lambda v) = \lambda \sigma(v)$$

• Try to find the geometric meaning